Yang Luo

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Stuttgart | Shanghai

Summary

- Hands-on experience in AI, robotics, and human-machine interaction, with strengths in prototyping and system integration.
- Experienced in turning complex requirements into practical product and technical solutions.

Skills

- **Software & Data:** Python | C# | Git | Power BI | MongoDB
- **AI & ML:** PyTorch | OpenCV | RAG | Generative AI
- **Design & Prototyping:** Grasshopper | Rhino | Universal Robots (UR)

Education

University of Stuttgart 2024.10 - 2026.10
- Integrative Technologies and Architectural Design Research — Stuttgart, Germany

Zhejiang University 2017.09-2022.06
- Bachelor of Architecture — Hangzhou, China

Professional Experience

STRABAG — 2025.03 - Present
Working Student (Data Services) — Stuttgart, Germany

Product Analytics | Product Ownership

- Designed and maintained a Power BI analytics dashboard used by 40+ internal stakeholders to analyze a production application serving hundreds of users for over 6 months.
- Built automated ***Python ETL*** workflows from ***MongoDB*** to structured analytical datasets.
- Collaborated with product and engineering teams to define analytics requirements, data sources, and user behavior metrics.

RoboticPlus.AI (Shanghai) Co., Ltd — 2022.08 - 2024.09
Solution Engineer — Shanghai, China

Robotics Application | System Integration | Solution Engineering

- Developed and integrated robotic fabrication workflows using 6-axis industrial robots, vision systems and custom software tools.
- Delivered robotics solutions from technical concept and prototyping to deployment, documentation and user training.

Key Projects:

Computational Design & Digital Fabrication Course(2023) of Zhejiang University
- Designed and delivered an 18-week robotic fabrication course covering computational design, industrial robotics, and robotic 3D printing.

End-to-End Robotic Fabrication of a Flexible Truss System -- Tongji University
- Collaborated with clients on laboratory solutions, including system layout, customized hardware, and automated fabrication workflows.
- Led process development and testing through 50+ iterations of workflow optimization.
- Established a streamlined fabrication workflow enabling automation and adaptability.

Robotic Simulation Plug-in -- RobimGH
- Contributed to the development and maintenance of RobimGH, a C#-based robotic arm planning plugin for Grasshopper, released on Food4Rhino.

Projects

Master Thesis Project

Robotic Assistance System for Proactive Human-Robot Collaboration

Individual Focus: Interaction System | Communication and System Integration

- Developed a ***human-centered robotic assistance system*** that detects human task states and proactively offers robot support during collaborative assembly.
- Designed the interaction loop from ***task perception → assistance decision → permission-based communication → robot execution → execution control***.
- Built a ML-based perception module for task-step and progress estimation from human motion data.
- Extended an existing rosbridge-based framework with event-driven HRI logic for ***human permission handling and execution control.***

link: https://github.com/Lo122/hrc_communication | https://github.com/Lo122/LSTM_HRC

Seminar Project

GenGarden: Human–AI–Robot Co-Creation

Individual Focus: Generative AI | Multi-Agent Workflow | HRC

- Developed an ***AI-enabled robotic co-creation system*** that integrated image generation, robotic execution, and visual evaluation into a physical sandbox design workflow.
- Built the system MVP and end-to-end workflow connecting camera input, a Gemini-based generation agent, Grasshopper, UR robot control, and execution evaluation agent.
- Designed a ***generate → execute → evaluate → rework*** process, with structured AI outputs translated into robot-executable data.

link: https://github.com/Lo122/zen_garden

Other Skills

- **Language:** Mandarin (native) | English (fluent)

Publication

CAADRIA 2022:

Guo, Y., **Luo, Y**., Wang, S., Tan, Y.Y., & Tracy, K. Robotic Fabrication of Topology Optimised Concrete Components with Reusable Formwork.